

Code:-

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <unistd.h>
4 #include <sys/types.h>
5 #include <sys/wait.h>
6 #include <string.h>
7
8 int main()
9 {
10     char command[100];
11     char *args[20];
12     char *token;
13     int i = 0;
14
15     printf("Enter a Linux command: ");
16     fgets(command, sizeof(command), stdin);
17
18     // Remove newline
19     command[strcspn(command, "\n")] = '\0';
20
21     // Split command into arguments
22     token = strtok(command, " ");
23
24     while (token != NULL && i < 19)
25     {
26         args[i++] = token;
27         token = strtok(NULL, " ");
28     }
29
30     args[i] = NULL;
31
32     pid_t pid = fork();
33
34     if (pid < 0)
35     {
36         perror("fork failed");
37         return 1;
38     }
39
40     if (pid == 0)
41     {
42         // Child process
43         printf("\nChild Process\n");
44         printf("Child PID : %d\n", getpid());
45         printf("Parent PID : %d\n", getppid());
46
47         execvp(args[0], args);
48
49         // Executes only if execvp fails
50         perror("execvp failed");
51         exit(1);
52     }
53     else
54     {
55         // Parent process
56         printf("\nParent Process\n");
57         printf("Parent PID : %d\n", getpid());
58         printf("Child PID : %d\n", pid);
59
60         wait(NULL);
61
62         printf("\nChild process completed.\n");
63     }
64
65     return 0;
66 }
```

Output:

```
Enter a Linux command: ls

Parent Process
Child Process
Parent PID : 565
Child PID : 566
Child PID : 566
Parent PID : 565
Experiment1  Practical15.c  Practical11  Practical13.c  Practical16  Practical3.c  Practical6  Practical8.c  Program16
              Program5
Experiment1.c Practical1  Practical11.c Practical14  Practical16.c Practical4  Practical6.c Practical19  Program16.c  Program5.c
Skill12.c    exp1.c
OS           Practical1.c Practical12  Practical14.c Practical2  Practical4.c Practical7  Practical9.c Program2.c
              ProgrammingPractical SurendraDesktopOperating
OSSP         Practical10  Practical12.c Practical15  Practical2.c Practical5  Practical7.c Program1  Program3.c
              Skill1      Systems
Operating    Practical10.c Practical13  Practical15.c Practical3  Practical5.c Practical8  Program1.c Program4.c
              Skill1.c      and
              snap

Child process completed.
```